

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Real Numbers

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the HCF and LCM of 84 and 126.

Answer:

$$\text{HCF} = 42. \text{ LCM} = (84 \times 126)/42 = 252.$$

Question 2

Find the HCF and LCM of 96 and 144.

Answer:

$$\text{HCF} = 48. \text{ LCM} = (96 \times 144)/48 = 288.$$

Question 3

Find the HCF and LCM of 135 and 225.

Answer:

$$\text{HCF} = 45. \text{ LCM} = (135 \times 225)/45 = 675.$$

Question 4

Find the HCF and LCM of 156 and 234.

Answer:

$$\text{HCF} = 78. \text{ LCM} = (156 \times 234)/78 = 468.$$

Question 5

Find the HCF and LCM of 108 and 180.

Answer:

$$\text{HCF} = 36. \text{ LCM} = (108 \times 180)/36 = 540.$$

Question 6

Find the HCF and LCM of 72 and 168.

Answer:

$$\text{HCF} = 24. \text{ LCM} = (72 \times 168)/24 = 504.$$

Question 7

Find the HCF and LCM of 175 and 245.

Answer:

$$\text{HCF} = 35. \text{ LCM} = (175 \times 245)/35 = 1225.$$

Question 8

Find the HCF and LCM of 252 and 378.

Answer:

$$\text{HCF} = 126. \text{ LCM} = (252 \times 378)/126 = 756.$$

Question 9

Find the HCF and LCM of 210 and 280.

Answer:

$$\text{HCF} = 70. \text{ LCM} = (210 \times 280)/70 = 840.$$

Real Numbers - Continued

Question 10

Find the HCF and LCM of 144 and 360.

Answer:

$$\text{HCF} = 72. \text{ LCM} = (144 \times 360)/72 = 720.$$

Question 11

Without division, decide whether $7/40$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 40. After removing factors 2 and 5, the remaining factor is 1; therefore the decimal expansion is terminating.

Question 12

Without division, decide whether $13/125$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 125. After removing factors 2 and 5, the remaining factor is 1; therefore the decimal expansion is terminating.

Question 13

Without division, decide whether $11/90$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 90. After removing factors 2 and 5, the remaining factor is 9; therefore the decimal expansion is non-terminating recurring.

Question 14

Without division, decide whether $17/800$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 800. After removing factors 2 and 5, the remaining factor is 1; therefore the decimal expansion is terminating.

Question 15

Without division, decide whether $29/64$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 64. After removing factors 2 and 5, the remaining factor is 1; therefore the decimal expansion is terminating.

Question 16

Without division, decide whether $3/70$ has a terminating decimal expansion.

Answer:

In lowest form the denominator is 70. After removing factors 2 and 5, the remaining factor is 7; therefore the decimal expansion is non-terminating recurring.

Question 17

Use Euclid's division algorithm to write 455 in the form $42q + r$.

Answer:

$$455 = 42 \times 10 + 35, \text{ where } 0 \leq 35 < 42.$$

Real Numbers - Continued

Question 18

Use Euclid's division algorithm to write 867 in the form $23q + r$.

Answer:

$867 = 23 \times 37 + 16$, where $0 \leq 16 < 23$.

Question 19

Use Euclid's division algorithm to write 1001 in the form $36q + r$.

Answer:

$1001 = 36 \times 27 + 29$, where $0 \leq 29 < 36$.

Question 20

Use Euclid's division algorithm to write 391 in the form $17q + r$.

Answer:

$391 = 17 \times 23 + 0$, where $0 \leq 0 < 17$.

Question 21

State the main formula, rule or theorem used in Real Numbers.

Answer:

For positive integers a and b : $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$.

Question 22

What do the important symbols or quantities in Real Numbers represent?

Answer:

A rational number p/q is terminating only if reduced q has prime factors 2 and/or 5.

Question 23

Write the correct method for solving a Real Numbers numerical.

Answer:

Use prime factorisation or Euclid's algorithm, then apply the relevant rule.

Question 24

Which unit, sign convention or condition must be checked in a Real Numbers answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Real Numbers be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Real Numbers questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Real Numbers - Continued

Question 27

How should full working be shown in a Real Numbers numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Real Numbers?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Question 29

How can a student decide which formula to use in Real Numbers?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Real Numbers?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: CBSE Mathematics curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.

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